Tutorial 6 (Acceleration Analysis)

(***Analytical method is to be used by default; the descriptions of the problems in the book are refined***)

- **Problem 7-4**
  The link lengths, coupler point location, and the values of θ_2, ω_2, and α_2 for the same fourbar linkages as used for position and velocity analysis in Chapters 4 and 6 are redefined in **Table P7-1**, which is basically the same as **Table P6-1**. The general link-age configuration and terminology are shown in **Figure P7-1**.

  <u>*For **Row a** assigned,*</u> based on the solutions for **Problem 4-7** in **Tutorial 3** and **Problem 6-5** in **Tutorial 5**, and only for the open configuration of the linkage, calculate the acceleration of point *A,* angular acceleration $\alpha3$ and $\alpha4$, and then calculate the accelerations of points *B* and *P*. Sketch the vectors of the accelerations of points A, B and P, showing their magnitudes and directions.

- **Problem 7-6**
  The link lengths and offset and the values of θ_2, ω_2, and α_2 for some noninverted, offset fourbar crank-slider linkages are defined in **Table P7-2**. The general linkage configuration and terminology are shown in **Figure P7-2**.

  <u>*For **Row a** assigned*</u>, based on the solutions for **Problem 4-10** in **Tutorial 3** and **Problem 6-7** in **Tutorial 5**, and only for the open configuration of the linkage , calculate the accelerations of the pin joints *A* and *B* and the acceleration of slip at the sliding joint.